
Collections of Things

Ordered List

An ordered list is a collection of things where they come in order. If it's not empty, then one and only one comes first. An ordered set can be accessed by an index like a list in a computer program.

Two ordered lists are the same if they have the same number of elements and if for each index the corresponding element is the same.

These are different ordered lists.

```
Out[ ]={
{5, 1, 0, 4, 2, 3, 6, 1, 2, 7}
{6, 2, 1, 4, 5, 1, 3, 0, 7, 2}
{1, 1, 5, 6, 4, 0, 7, 2, 3, 2}
{6, 4, 1, 1, 7, 3, 0, 5, 2, 2}
```

Generally, it is allowed to have repeated elements in an ordered list.

It is allowed to have an empty list: {}

Vectors

In the special case where the elements are numbers, the ordered list is called a vector.

The length is the dimension.

Theoretically, it is convenient to have a zero dimensional vector space, which has one element {}, although in practice the dimension is almost always a positive integer.

Matrices

In the special case where the elements are vectors, and they all have the same dimension, the ordered list is called a matrix.

The dimensions of a matrix is its rows and columns, which don't need to be the same.

Tensors

Arrays of higher rank can be defined recursively. If vectors are rank 1, and matrices are rank 2, then a tensor of rank k is an ordered list of tensors of rank $k - 1$, where each element has the same dimensions.

Unordered List

In an unordered list, the order doesn't matter.

These are the same as unordered lists.

```

{1, 6, 2, 2, 3, 4, 0, 1, 7, 5}
{6, 1, 1, 2, 4, 0, 2, 7, 3, 5}
Out[ #]= {4, 2, 5, 2, 1, 3, 7, 0, 6, 1}
{2, 3, 0, 4, 1, 7, 5, 2, 6, 1}

```

One way to check if two unordered lists are the same is to pretend they are ordered lists and then sort the elements. Then check if the sorted ordered lists are the same.

```

{0, 1, 1, 2, 2, 3, 4, 5, 6, 7}
{0, 1, 1, 2, 2, 3, 4, 5, 6, 7}
Out[ #]= {0, 1, 1, 2, 2, 3, 4, 5, 6, 7}
{0, 1, 1, 2, 2, 3, 4, 5, 6, 7}

```

Unordered List without repeats = Set

A set is an unordered list with no repeated elements.

One imagines a dataset with no repeats, where the entries could have gone in any order.

For instance, each row is a person and their vote for their favorite ice cream.

There are no repeats because each person only gets one vote.

The order doesn't matter because the order of the data collection doesn't affect the vote count.

If an unordered list has repeated elements it is not a set.

```

{5, 6, 0, 4, 7, 1, 3, 2}
{5, 7, 6, 1, 4, 3, 2, 0}
Out[ #]= {0, 7, 6, 2, 1, 5, 3, 4}
{4, 5, 6, 7, 2, 1, 0, 3}

```

We can check these are the same after sorting.

```

{0, 1, 2, 3, 4, 5, 6, 7}
{0, 1, 2, 3, 4, 5, 6, 7}
Out[ #]= {0, 1, 2, 3, 4, 5, 6, 7}
{0, 1, 2, 3, 4, 5, 6, 7}

```

If S is a set then either $x \in S$ or $x \notin S$ where x is anything that conceivably could be in S .

By ordering all possible x 's, a set can be represented by a list of True or False, or equivalently, 1 or 0.

Those are called boolean values. In this way operations on sets can be considered operations in a Boolean algebra.

An unordered set with repeats is called a **multiset**.

Direct Product or Cartesian Product

One way to produce a new set from two sets is the Cartesian product

$A \times B$ = the set of ordered pairs $\{a, b\}$ where $a \in A$ and $b \in B$

Note that $A = B$ is possible, for instance when you use Cartesian coordinates in the plane (x, y) where x and y can be any real number.

In general, the Cartesian product of n sets A_i is a set of ordered lists where the i -th indexed element is in

A_i

For finite sets, the number of elements in $A \times B$ is the number in A times the number in B .

It is possible to display the elements as a grid in the case of two sets.

Here $A = \{1, 2\}$ and $B = \{x, y, z\}$. Note $A \times B$ has $2 \times 3 = 6$ elements.

```
Out[4]=
{1, x} {1, y} {1, z}
{2, x} {2, y} {2, z}
```

As a set though it also makes sense to list the elements.

```
Out[5]= { {1, x}, {1, y}, {1, z}, {2, x}, {2, y}, {2, z} }
```

A set S can be realized by a Cartesian product if there are functions $f_1: S \rightarrow A$ and $f_2: S \rightarrow B$ that are suitable projections. They have to be onto and the preimages $f_1^{-1}(a) \cap f_2^{-1}(b)$ have to be a single element in S . There are other equivalent criteria.

Membership

x is an element of S if x is in the collection of things that form S .

Disjoint Sets

Two sets A and B are said to be disjoint if they have no elements in common.

What is allowed in a Set?

The elements could be primitives like numbers or strings or possible properties of a piece of data.

The elements can be sets. For instance, $S =$ the set of arithmetic sequences. Then if $x \in S$, it follows that $x = \{a n + b\}$ for some integers a and b . The set $\{ \dots, -2, 0, 2, 4, 6, 8, \dots \}$ is an element of S .

The elements could be ordered lists. It can be a mixture.

The number of elements can be finite or infinite.

Care must be taken. Naive set theory is a vague term that means that sets can be used in mathematics as long as they are ordinary in some sense. If one allows any description of a set to count as a set, then a paradox can cause problems.

Subsets

A is a subset of B if every element of A is also in B .

$A \subset B$

By this definition, S is a subset of S .

Caution: some authors write $A \subseteq B$ to indicate that A could be B , and reserve the above notation for A is a subset that is not equal to B .

Empty Set

The empty set $\{\}$ has no elements.

Often denoted by $\emptyset$

The empty subset is always a subset of S .

Power Sets

The power set $P(S)$ = the set of subsets of S .

The empty set is always in $P(S)$ and S is always in $P(S)$

For example, if $S = \{1,2,3,4\}$ this is $P(S)$

Out[*n*]= $\{\{\}, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\},$
 $\{2, 4\}, \{3, 4\}, \{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \{1, 2, 3, 4\}\}$

Because x is either in or not in, $P(S)$ has 2^n elements where $n = |S|$

For example, $P(\emptyset) = \{\{\}\}$ has 1 element.

Iterating the power set operation gives sets of size 2^n , which is one way to theoretically construct sets of all sizes.

Subsets defined by a property

$\{x \in S \mid \text{property } P(x) \text{ holds}\}$

One is allowed to construct sets of the above form, which are subsets of S , as long as the property P is well-defined.

Depending on the context, the property P does not need to be finitely computable.

Sometimes it is allowed to write a set defined by a property in a way that doesn't fit this pattern, but it's ok because it could be written this way.

For instance, the set of perfect squares

$\{n^2 \text{ such that } n \in \mathbb{Z}\}$

is acceptable, because it could be written more obscurely as

$\{m \in \mathbb{Z} \mid \exists n \in \mathbb{Z} \text{ such that } m = n^2\}$

Infinite Sets

Sets in Mathematics

The existence of these sets are either taken for granted, or in some sense, they are proven to exist.

natural numbers $\mathbb{N}$

integers $\mathbb{Z}$

rational numbers $\mathbb{Q}$

real numbers $\mathbb{R}$

complex numbers $\mathbb{C}$

An infinite set S is called countable if there is a 1-1 map $\mathbb{N} \rightarrow S$

Some sets are countable, even if not in an immediately obvious way, like $\mathbb{Q}$

Some sets are uncountable like $\mathbb{R}$

Set Operations

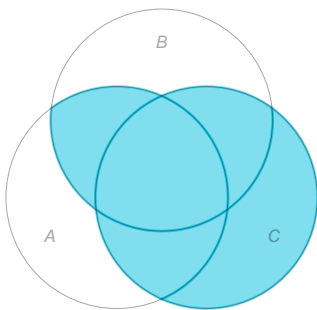
Certain expressions and operators are allowed with sets.

Venn diagrams

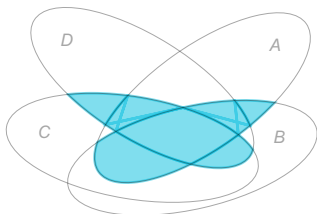
Set expressions can be visualized in a Venn diagram, as long as there are two or three sets involved.

Theoretically four or five is possible but not as useful for visualization.

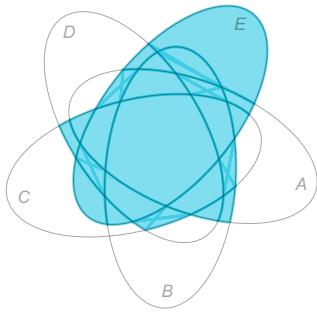
This is $(A \cap B) \cup C$ from Wolfram Alpha



This is $(A \cap B) \cup (C \cap D)$ from Wolfram Alpha



This is $(A \cap B) \cup (C \cap D) \cup E$ from Wolfram Alpha



A set expression defined using n sets is specified by 2^n boolean variables, so the Venn diagrams contain 2^n regions (including the outside).

Union

$A \cup B$ = everything in A plus everything in B

Intersection

$A \cap B$ = everything in both A and B

Two sets A and B are disjoint iff $A \cap B = \emptyset$

Complement

A complement = Everything not in A

This only makes sense when a set is inside of a set that can be considered everything.

Denoted by A^c or by $\bar{A}$

Period 2: $(A^c)^c = A$

Set difference

$A - B$ is everything in A that is not in B

Symmetric Difference

$A \oplus B$ is everything in A or B but not in both.

Boolean Algebra

There is a direct correspondence of set formulas with logical formulas.

Consider x as a possible element of the sets.

Then for each set A_i in a set formula $f(A_1, A_2, \dots, A_n)$, there is a truth value p_i , T or F, as follows

p_1 = is x in A_1 ?

$p_2 = \text{is } x \text{ in } A_2?$

....

$p_n = \text{is } x \text{ in } A_n?$

Then union = $\cup$ corresponds to Or

intersection = $\cap$ corresponds to And

complement corresponds to Not

Set difference corresponds to Xor

For instance, if $f(A_1, A_2) = A_1 - A_2$ then x would be in A_1 and not in A_2 so we get the boolean expression

$$p_1 \wedge (\neg p_2)$$

Set Identities

Because the set formulas correspond to boolean expressions, we get the same identities.

Note that as in the case of boolean expressions with Or and And, there is a formal symmetry between Union and Intersection.

$(p \cap q)^c = p^c \cup q^c$	De Morgan
$(p \cup q)^c = p^c \cap q^c$	De Morgan
$p = p \cap p$	idempotent of and
$p = p \cup p$	idempotent of or
$p \cap q = q \cap p$	commutativity of and
$p \cup q = q \cup p$	commutativity of or
$p = (p^c)^c$	law of double negation
$p = p \cap (p \cup q)$	adsorption
$p = p \cup (p \cap q)$	adsorption
$p \cap p^c = q \cap q^c$	law of noncontradiction
$p \cup p^c = q \cup q^c$	law of excluded middle
$p \cap (q \cap r) = (p \cap q) \cap r$	associativity of and
$p \cup (q \cup r) = (p \cup q) \cup r$	associativity of or
$p \cap (q \cup r) = (p \cap q) \cup (p \cap r)$	first distributive law
$p \cup (q \cap r) = (p \cup q) \cap (p \cup r)$	second distributive law

Out[13]=

Functions

Consider a function from set X to set Y .

Denoted by $f: X \rightarrow Y$

For each $x \in X$, there is a unique $y \in Y$ such that $y = f(x)$.

That's the vertical line test.

We allow the possibility that not everything in Y is covered. For instance, $f(x) = \cos(x)$ is never 2.

Image in the Range

The image of f are all the y values covered by the function.

$$\{y \in Y \mid \exists x \in X, y = f(x)\}$$

For $A \subset X$, the image of A is $f(A) \subset Y$

Preimage

The preimage of $B \subset Y$ is the $x \in X$ where $f(x) \in B$

Denoted f^{-1} even if this is not a function or an actual inverse.

For $B \subset Y$, the preimage is $f^{-1}(B) \subset X$

Injective or one-to-one

A function is 1-1 if whenever $f(x_1) = f(x_2)$ it follows that $x_1 = x_2$

Also called injective or one-to-one.

Every value of y is hit once or not at all.

Surjective or onto

A function is onto if every value of y is hit.

$$\forall y \in Y, \exists x \in X \text{ such that } y = f(x)$$

A function can be onto without being 1-1, like $f: [0, \pi] \rightarrow [-1, 1]$ with $f(x) = \cos(x)$. Every value in $Y = [-1, 1]$ is a cosine of some angle. This function is not 1-1 because $f(0) = f(\pi)$ and both 0 and π are in X .

A function can be 1-1 without being onto like $f: [0, 1] \rightarrow [0, \pi/2]$ with $f(x) = \arctan(x)$. Each number in $X = [0, 1]$ has a unique angle where $y = \arctan x$, but not all angles are covered. Only the subset $[0, \pi/4]$ in this case.

Bijjective or invertible

A bijective function is 1-1 and onto.

A bijective function has an inverse function, which is also a bijection.

For each x there is a unique y , and for each y there is a unique x .

Out[$\ast$]=

x	p (x)
1	1
2	4
3	2
4	3

x	q (x)
1	1
2	3
3	4
4	2

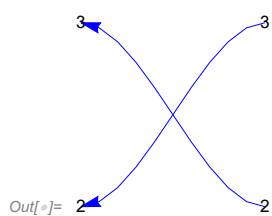
Permutation

In the case of $f: X \rightarrow X$, if f is a bijection then it is a **permutation**.

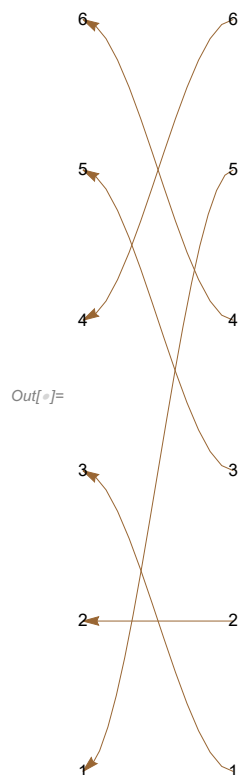
In[14]:= `Column[Permutations[{a, b, c}]]`

Out[14]=
`{a, b, c}`
`{a, c, b}`
`{b, a, c}`
`{b, c, a}`
`{c, a, b}`
`{c, b, a}`

a size 3 permutation

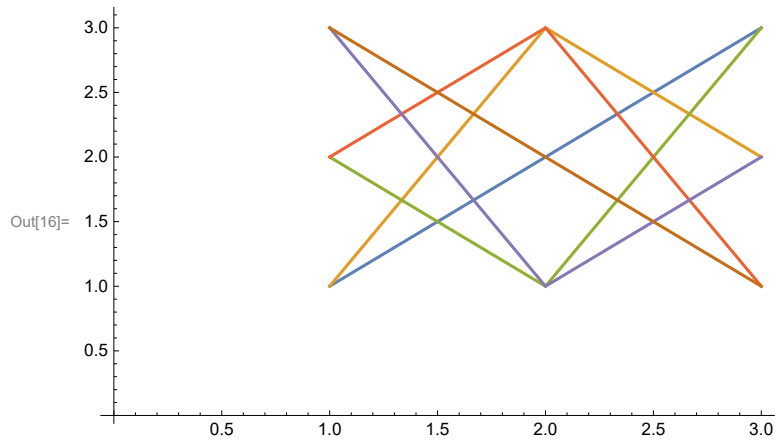


a size 6 permutation



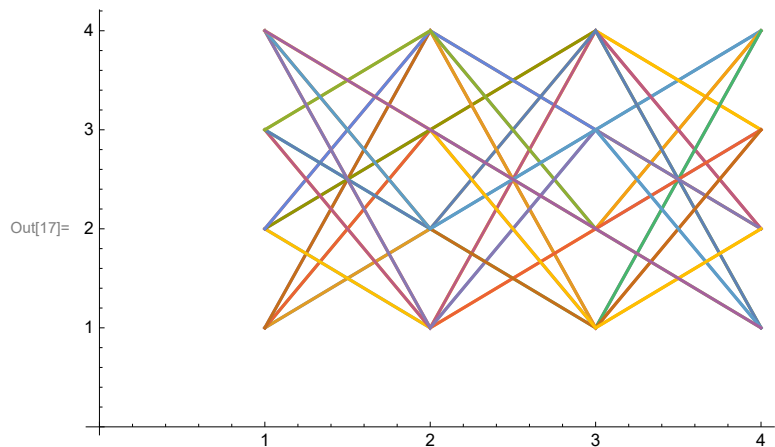
We can try plotting all permutations. Here permutations of $\{1,2,3\}$

In[16]:= **ListLinePlot** [Permutations [Range [3]]]



and here the permutations of $\{1,2,3,4\}$

In[17]:= **ListLinePlot** [Permutations [Range [4]]]



How many permutations are there if X has n elements?

Composition of Permutations

The composition of permutations is also a bijection because it is a 1-1 map and onto. [Why?]

Since the composition of functions is associative, the composition of permutations is associative.

The composition $p \cdot q$ is defined by q going first. So $p \cdot q(x) = p(q(x))$. This is the convention in our textbook, but other textbooks will have other conventions.

Example: Let $S = \{1, 2, 3\}$ and p and q are given by the following tables.

Out[18]=

x	p (x)
1	1
2	3
3	2

x	q (x)
1	2
2	1
3	3

Here are the two cases $q \cdot p$ and $p \cdot q$

Out[]=

x	p (x)	q (p (x))
1	1	2
2	3	3
3	2	1

x	q (x)	p (q (x))
1	2	3
2	1	1
3	3	2

For example, $q \cdot p(2) = q(3)$ because $p(2) = 3$.

The first thing to notice is that the order matters. The group of all permutations is not abelian.

Discussion

Most maps are not onto and not one-to-one

Out[]=

n	fraction of bijective maps in all maps from n things to n things	percent
1	1	100%
2	$\frac{1}{2}$	50%
3	$\frac{2}{9}$	22. %
4	$\frac{3}{32}$	9.4%
5	$\frac{24}{625}$	3.8%
6	$\frac{5}{324}$	1.5%
7	$\frac{720}{117\,649}$	0.61%

How can we calculate this?

Enumeration

An enumeration of a set X is a bijective map $f: \{1, 2, \dots, n\} \rightarrow X$

Using functions for counting

Often one wants to know how many elements are in X .

One way to count in a rigorous way is to have a well defined enumeration.

How many ordered pairs in the Cartesian product $\{1, 2, 3\}^n$?

The map doesn't need to be a bijection to be useful for counting.

How many integers n when squared are between 1 and 100 inclusive? When $n^2 > 0$ there is a positive root and a negative root, so we count the numbers in the image and multiply by 2, or $10(2) = 20$.